Nightrol™ - Official Integration Guide

1. Introduction

Thank you for choosing **Nightrol™ - Simple Secure Save**.

This module is designed so that even beginner Unity developers can build a **secure game save system in just 10 seconds**.

With **AES-256 encryption** and **HMAC integrity verification**, it protects valuable game data from unauthorized modification or hacking.

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2. Quick Setup

You can complete the basic setup **without coding** using the editor's top menu:

1. **Create Manager**

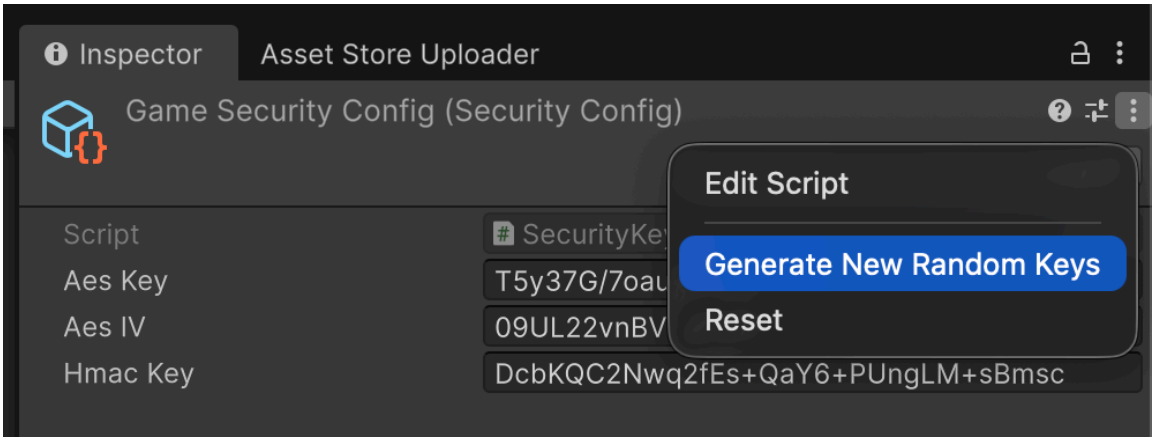
   Click the top menu `Tools` → `Nightrol™` → `Create GameDataManager`.

   - A `GameDataManager` object will automatically be created in the Scene.
   - Make sure the `GameDataManager` script is attached to the `GameDataManager` object component.
   - If it is missing, manually add the `GameDataManager` script using **Add Component**.

2. **Generate Security Key**

   Click the top menu `Tools` → `Nightrol™` → `Make Key File`.

   - A `GameSecurityConfig` ScriptableObject will be created in the Project.
   - A random key will be generated automatically upon creation.
   - If the key is not generated randomly or you want to change it, press the button below to regenerate it.



Note: The keys should preferably use the following byte sizes:

Key Type	Size
AES Key	16 bytes
AES IV	16 bytes
HMAC Key	32 bytes

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3. Main Features & Code Usage

Saving Data

Encrypts and saves the current data stored in `GameDataManager` to the local storage.

```
GameDataManager.Instance.Save();
```

Customizing the Data Structure

Simply open the `GameData.cs` file and add the variables you want.

They will automatically be included in the encrypted save data.

Examples:

- `PlayerData` : Level, experience, health, etc.
- `SettingsData` : Volume, graphics settings, etc.

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4. Security Technology

- **AES Encryption**

Data is encrypted into an unreadable binary format through `AESSecurityHelper`.

- **HMAC Integrity**

  `HMACHelper` generates a **signature (fingerprint)** of the data.

  If a user modifies even a single value using a text editor or other tool, the integrity verification will fail, preventing corrupted or tampered data from being used.

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5. FAQ

Where is the save file located?

- It uses Unity's `Application.persistentDataPath`.
- The exact file path is displayed when the game runs.

What happens if data tampering is detected?

- A **red error log** will appear in the console.
- For safety, the system will automatically **reset to a new default game data file**.

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6. Support & Review

If this asset helped you, please consider leaving a **5-star review on the Asset Store** and starring the project.

Your support helps us continue improving and updating the asset.

- **GitHub**: https://github.com/PotentialProject/Nightrol
- **Publisher**: Nightrol™